In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, reca

df = pd.read_csv(r"E:\diabetes.csv")

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)
print(df.head())

print("\nMissing values:")
print(df.isnull().sum())

X = df.drop("Outcome", axis=1)
y = df["Outcome"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining data shape:", X_train.shape)
print("Testing data shape:", X_test.shape)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

k = 5

knn = KNeighborsClassifier(n_neighbors=k)
knn.fit(X_train_scaled, y_train)

y_pred = knn.predict(X_test_scaled)

cm = confusion_matrix(y_test, y_pred)
accuracy = accuracy_score(y_test, y_pred)
error_rate = 1 - accuracy
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

print("\nAccuracy:", accuracy)
```

```
print("Accuracy (%):", accuracy * 100)

print("\nError Rate:", error_rate)
print("Error Rate (%):", error_rate * 100)

print("\nPrecision:", precision)

print("\nRecall:", recall)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nKNN Results")
print("K =", k)
print("Accuracy =", round(accuracy, 4))
print("Error Rate =", round(error_rate, 4))
print("Precision =", round(precision, 4))
print("Recall =", round(recall, 4))

plt.figure(figsize=(6, 5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Diabetes", "Diabetes"],
    yticklabels=["No Diabetes", "Diabetes"]
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("KNN Confusion Matrix")
plt.show()
```

```
Dataset loaded successfully!
Dataset shape: (768, 9)
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   Pedigree  Age  Outcome
0     0.627   50        1
1     0.351   31        0
2     0.672   32        1
3     0.167   21        0
4     2.288   33        1

Missing values:
Pregnancies      0
Glucose          0
BloodPressure    0
SkinThickness    0
Insulin          0
BMI              0
Pedigree         0
Age              0
Outcome          0
dtype: int64

Training data shape: (614, 8)
Testing data shape: (154, 8)

Confusion Matrix:
[[80 20]
 [26 28]]

Accuracy: 0.7012987012987013
Accuracy (%): 70.12987012987013

Error Rate: 0.2987012987012987
Error Rate (%): 29.87012987012987

Precision: 0.5833333333333334

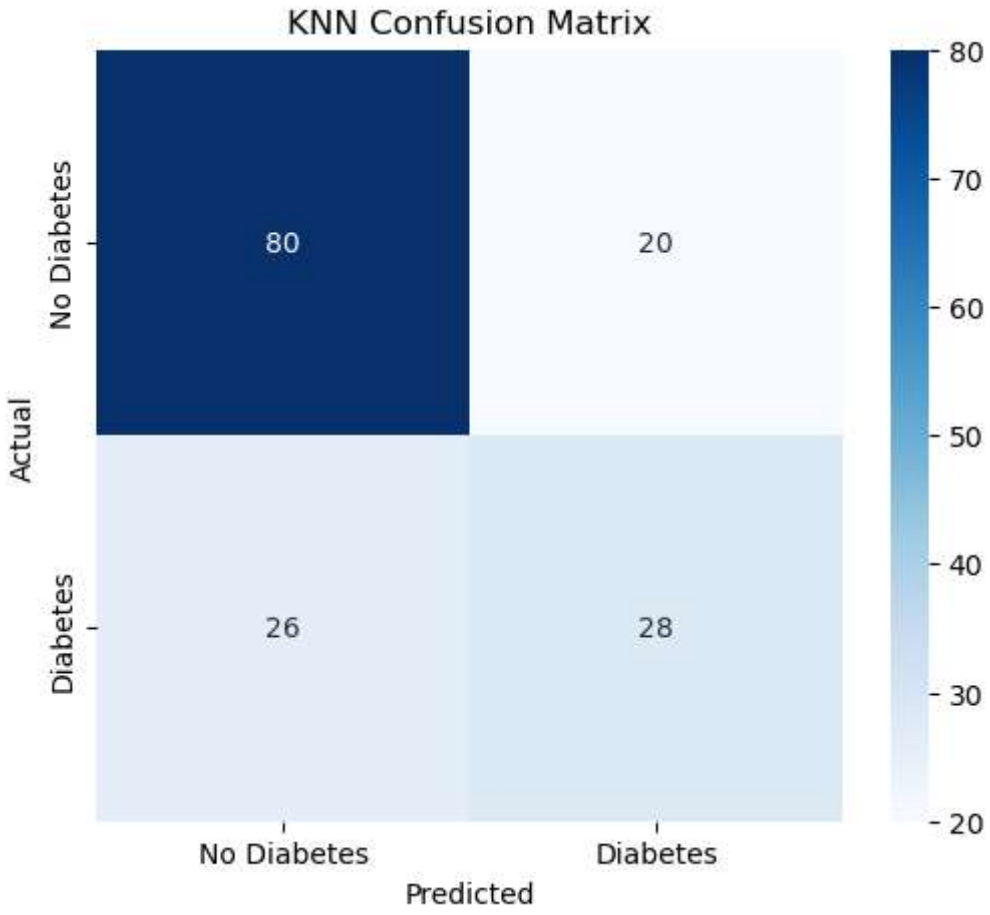
Recall: 0.5185185185185185

Classification Report:
              precision    recall  f1-score   support

           0       0.75      0.80      0.78       100
           1       0.58      0.52      0.55        54

    accuracy                           0.70       154
   macro avg       0.67      0.66      0.66       154
weighted avg       0.69      0.70      0.70       154
```

```
KNN Results
K = 5
Accuracy = 0.7013
Error Rate = 0.2987
Precision = 0.5833
Recall = 0.5185
```



In [ ]: